

Journal section: Operative Dentistry and Endodontics
Publication Types: Research

doi:10.4317/jced.57902
<https://doi.org/10.4317/jced.57902>

A comparative evaluation of Advanced Platelet-Rich Fibrin (A-PRF) and Platelet-Rich Fibrin (PRF) as a Scaffold in Regenerative Endodontic Treatment of Traumatized Immature Non-vital permanent anterior teeth: A Prospective clinical study

Veena Jayadevan ¹, Paras-Mull Gehlot ², Vinutha Manjunath ², Subbarao V. Madhunapantula ³, Jyothi-Swandenahalli Lakshmikanth ⁴

¹ BDS. Department of Conservative Dentistry and Endodontics. JSS Dental College and Hospital. JSS Academy of Higher Education and Research. Sri Shivarathreeshwara Nagar. Mysuru- 570015, Karnataka, India

² MDS. Department of Conservative Dentistry and Endodontics. JSS Dental College and Hospital. JSS Academy of Higher Education and Research. Sri Shivarathreeshwara Nagar. Mysuru- 570015, Karnataka, India

³ BSc, B.Ed, M.Sc, Ph.D. Department of Biochemistry. JSS Medical College and Hospital. JSS Academy of Higher Education and Research. Sri Shivarathreeshwara Nagar. Mysuru- 570015, Karnataka, India

⁴ M Pharm. Department of Pharmaceutics. JSS College of Pharmacy. JSS Academy of Higher Education and Research. Sri Shivarathreeshwara Nagar. Mysuru- 570015, Karnataka, India

Correspondence:

*Department of Conservative Dentistry and Endodontics
JSS Dental College and Hospital
JSS Academy of Higher Education and Research
Sri Shivarathreeshwara Nagar
Mysuru- 570015, Karnataka, India
dr.parasmullj@jssuni.edu.in*

Jayadevan V, Gehlot PM, Manjunath V, Madhunapantula SV, Lakshmikanth JS. A comparative evaluation of Advanced Platelet-Rich Fibrin (A-PRF) and Platelet-Rich Fibrin (PRF) as a Scaffold in Regenerative Endodontic Treatment of Traumatized Immature Non-vital permanent anterior teeth: A Prospective clinical study. J Clin Exp Dent. 2021;13(5):e463-72.

Received: 27/09/2020
Accepted: 20/12/2020

Article Number: 57902 <http://www.medicinaoral.com/odo/indice.htm>
© Medicina Oral S. L. C.I.F. B 96689336 - eISSN: 1989-5488
eMail: jced@jced.es
Indexed in:
Pubmed
Pubmed Central® (PMC)
Scopus
DOI® System

Introduction

The management of an immature non-vital tooth is a challenge in Endodontics due to difficulty in performing chemo-mechanical debridement and creating an effective apical seal by conventional endodontic treatment (1). Pulpal necrosis frequently occurs due to dental trauma in children (2). Apexification has been suggested in the management of immature non-vital tooth, which can induce apical closure through the formation of mineralized tissue in the apical region. However, apexification does not increase the root dentin thickness nor increase the root length/apical closure which eventually could lead to the fracture of the tooth (3,4).

Recently, Regenerative endodontic therapy (RET) has been recommended in management of immature tooth with necrotic pulp and/or apical periodontitis/abscess (5). RET is a “biologically-based procedures designed to physiologically replace damaged tooth structures, including dentin and root structures, as well as cells of the pulp-dentin complex” (6). The key components of RET include: (i) Stem cells mainly from Apical papilla (SCAPs), (ii) Growth factors/morphogens and (iii) Scaffold that can support cell growth and differentiation (5). The ideal characteristics of a scaffold for successful regeneration have been summarized by Nosrat et al (7). The use of blood clot, platelet-rich plasma (PRP), and platelet-rich fibrin (PRF) as bioscaffolds has been reported in literature (8, 9). PRP is a natural scaffold that eliminates the risk of cross-infection and immunogenicity. However, it is not 100% autologous (10). PRF was introduced in RET to overcome this limitation of PRP and furthermore PRF also exhibited enhanced release of growth factors and leukocytes. With a better understanding of the use of autologous platelet concentrates and their role in healing, newer variants of PRF have been developed. Studies demonstrated that a slight increase in centrifugation time and reduction in speed resulted in the development of advanced platelet-rich fibrin (A-PRF), a variant of standard PRF with better regenerative potential, which is commonly used in periodontal regeneration and implant surgery (10,11).

A-PRF membrane have shown promising response in the endodontic surgery (12). However, the use of A-PRF as a scaffold in regenerative endodontics has not been

attempted. Thus, this prospective clinical study aimed to evaluate and compare the treatment outcomes of A-PRF and PRF as a scaffold in RET of traumatized immature non-vital teeth.

Material and Methods

This prospective clinical study was conducted according to the declaration of Helsinki (1975) regarding biomedical research involving human subjects and the protocol was approved by the institutional ethics committee.

The study was performed from April 2018 to September 2019 at a single Centre of Department of Conservative Dentistry and Endodontics. The study design was randomized with a two-arm, parallel design (1:1 allocation ratio). Regenerative endodontic treatments (RET) were performed either using A-PRF or PRF as a scaffold.

Patients reporting to the outpatient of the Department of Conservative Dentistry and Endodontics and Department of Pediatric Dentistry were enrolled. Patients aged between 8 and 30 years meeting the inclusion criteria were randomly allotted to the two groups of A-PRF (n=15) and PRF (n=15). Cooperative patients with an immature single-rooted non-vital tooth with apical width more than 1 mm, with or without periapical lesion, and trauma as an etiology, with no systemic diseases were included in the study. Teeth with attempted access cavity without intracanal medicament placement or two teeth per participant could be included in the study. Teeth with mature apex, internal or external resorption, mobility, ankylosis, root fracture, periodontal pocket, and unrestorable structure were excluded from the study. Furthermore, patients with known allergy to antibiotics and local anesthetic agents were also excluded from the study.

Participants were informed about the purpose, procedure, and expected outcome of the study along with alternate treatment options. A signed informed assent/consent form was obtained from each patient or parent.

-Clinical procedures

A preoperative diagnosis of pulp necrosis was established based on the electric pulp test (Parkell, Inc., NY, USA) and cold testing with Roeko Endo-Frost (Coltene/Whaledent GmbH, Germany). A two-visit RET was performed by a single operator. The clinical protocol for

RET procedure was in accordance with the American Association of Endodontists (AAE) guidelines (13,14). First visit:

Local anesthesia containing 2% lignocaine hydrochloride and 1:80,000 adrenaline (Lignox A, Indoco Remedies Ltd., Gujarat, India) was administered and access cavity prepared under rubber dam isolation. Pulpal remnants were extirpated using barbed broaches and copious, gentle irrigation with 1.5% sodium hypochlorite (NaOCl) (Nice Chemicals (P) Ltd., Kochi, India) (20 mL/canal, 5 min) using a 30-G side venting needle (RC Twents, Prime Dental. Pvt Ltd., Mumbai, India) about 2 mm short of the root end. The canal was then irrigated with saline (Denis Chem Lab Limited, Kalol, India) and 17% ethylenediaminetetraacetic acid (EDTA) (Canalarge, Ammdent, India) (20 mL/canal, 5 min). Radiographic working length was determined using paralleling technique (Gendex Visualix, Dentsply, Italy). Minimal canal debridement was done using K-files (#80–120 size, Mani Inc., Japan). Canal was dried using paper points (DiaDent, Korea), and modified triple antibiotic paste was placed in the canal till the working length and coronally short of the cemento-enamel junction (CEJ) using a lentulospiral. For preparation of the modified triple antibiotic paste (TAP), an equal proportion (1:1:1) of sugar-free tablets of ciprofloxacin, metronidazole (Albert David Limited, WB, India), and cefaclor (Health biotech limited, Chandigarh, India) was powdered and mixed with 1 ml of propylene glycol (vehicle) to obtain a stock solution of 100mg/ml, which was diluted to obtain a final concentration of 1mg/mL (8, 15). The access cavity was sealed with a temporary restorative material of at least 3–4 mm thickness (Cavit. 3M ESPE, St. Paul, MN, USA). Patients were recalled after 4 weeks.

Second Visit (After 4 weeks):

The patients were assessed for any signs or symptoms of persistent infection, and the intracanal medicament was repeated if required. Asymptomatic patients were recruited for RET.

Plain local anesthesia with 3% Mepivacaine (Scandonest, Septodont, Canada) without vasoconstrictor was administered. All the steps were performed using dental operating microscope (OPMI Pico, Carl Zeiss, Oberkochen, Germany). The tooth was re-accessed under rubber dam isolation. Copious and gentle irrigation with saline and 20 ml of 17% EDTA was performed to remove residue of the TAP, and paper points were used to dry the canal. Bleeding was induced in the canal system by overinstrumentation and rotation of a slightly precurved K-file (#40 size, Mani Inc., Japan) at 2 mm past the apical foramen with the aim of having the canal filled with blood to the level of the CEJ. A standard venipuncture was performed (median cubital vein). A-PRF or PRF was freshly prepared using a centrifuge (R-8C Laboratory centrifuge, Remi Lab, Mumbai, India). For PRF, 10

ml of intravenous blood was drawn into a tube without anticoagulant and centrifuged at 2700 rpm for 12 min (10). For A-PRF, 10 ml of intravenous blood was drawn into a tube without anticoagulant and centrifuged at 1500 rpm for 14 min (11).

The fibrin clots from both A-PRF and PRF were removed from the tubes and separated from the red element phase below the buffy coat using sterile scissors. The clot was delicately placed inside a sterile box (Magic PRF & GRF Box, Jull-Dent, Mumbai, India) and squeezed between sterile metal plates to obtain a membrane. The A-PRF or PRF clot was then placed into the root canal using an endodontic plugger. Biodentine (Septodont, Saint-Maur-des-Fossés, France) was placed as a capping material over the A-PRF or PRF. Glass ionomer cement (GIC) (GC Fuji IX, GC India) was placed gently in a thickness of about 3–4 mm over the Biodentine and the access was temporized with Cavit. A postoperative radiograph was taken to evaluate the placement of Biodentine and orifice barrier.

Patients were dismissed with instructions about the follow-up and for reporting in case of any symptoms. In cases where intracanal bleeding was not achieved, patients were placed into the rescue group to undergo conventional Biodentine apexification. Post regenerative treatment consisted of non-vital bleaching or composite restoration. These procedures were performed after a period of one week.

-Clinical and Radiographic evaluations

Patients were reviewed after 6 months (± 2 weeks) and 12 months (± 2 weeks) clinically, and radiographs were taken using the same x-ray device and paralleling technique by the same operator. During the 12-month follow-up period, 9 patients were excluded from the study. Three radiographic evaluations were done at the end of the 12 months of follow-up:

(i) Periapical status

The periapical healing was evaluated using the periapical index (PAI) given by Orstavik *et al.* (16). The PAI scores of the preoperative and postoperative radiographic images were assigned by a single investigator for all the cases.

Treatment outcomes were categorized as follows (17): (a) Healed: Both the clinical (subjective and objective) and radiographic presentations (PAI score 1 or 2) were normal (b) Healing: The periapical radiolucency was reduced (PAI score 3 or 4) with a normal clinical presentation and (c) Diseased: The radiolucency had either increased or persisted without change (PAI score increased or unchanged) even when the clinical presentation was normal, or the clinical signs or symptoms were present regardless of the radiographic presentation.

(ii) Determination of changes in the root length and dentin thickness.

The preoperative and postoperative root length and den-

tin thickness were measured using the open source software, ImageJ (National Institutes of Health, Bethesda, MD). The TurboReg plugin (Biomedical Imaging Group, Swiss Federal Institute of Technology, Lausanne, VD, Switzerland) was used with the software to minimize any dimensional change due to angulation errors in the preoperative and postoperative radiographs (18). The image sizes were calibrated to the size #2 of an intraoral radiographic film, which facilitated the measurement of changes in root size on a millimeter (mm) scale (18,19). The root length and thickness were measured using the “straight-line” tool of TurboReg. A modified protocol proposed by Alobaid *et al.* was applied for radiographic measurements (20). Accordingly, the measurements for root length were performed as a straight line from the CEJ to the midpoint of the apex of the root from both the mesial and distal points, and then both measurements were averaged to obtain the total root length (Fig. 1A).

asurements obtained by the two investigators was considered as the final value for each radiographic outcome.

(iii) Apical response of immature tooth:

The apical response of the immature tooth to RET was evaluated according to Chen *et al.* (5). Type 1: Increased thickening of the canal walls and continued root maturation. Type 2: No significant continuation of root development with the root apex becoming blunt and closed. Type 3: Continued root development with the apical foramen remaining open. Type 4: Severe calcification (obliteration) of the canal space. Type 5: Hard tissue barrier formed in the canal space between the coronal Biodentine plug and the root apex.

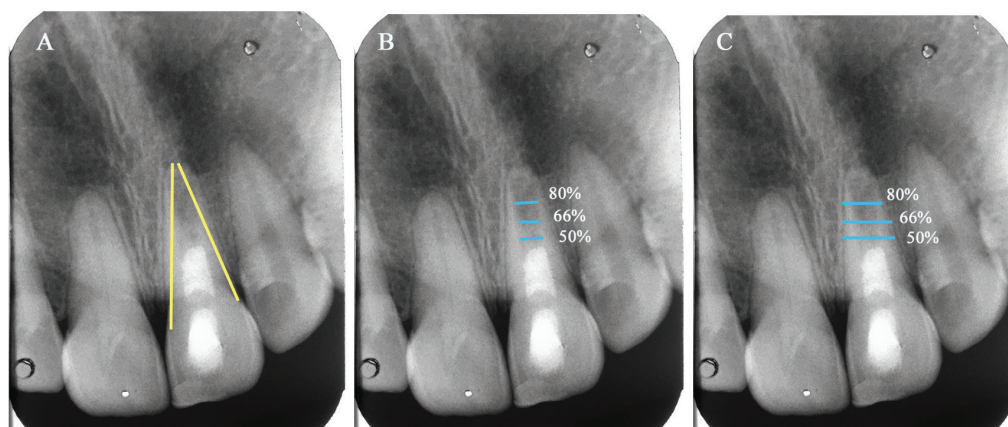
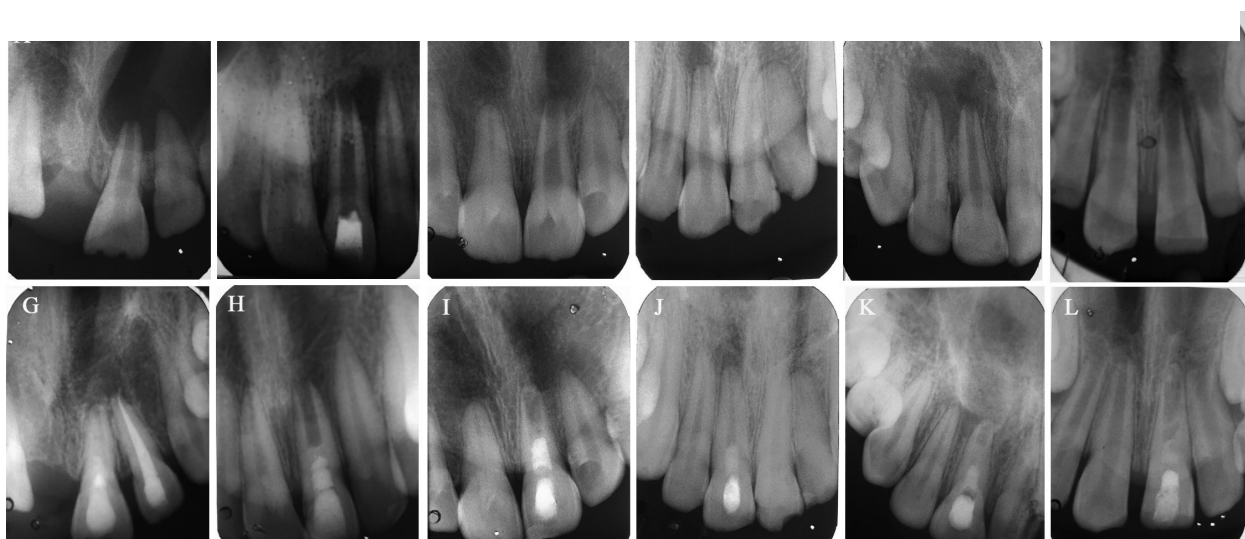


Fig. 1: Radiographic root measurement methods, using ImageJ software. (A) Root length measurement (B) pulp space width measurement, and (C) total root width measurement. The root thickness was determined subtracting pulp space width from the total root width.



Conclusions

The results of the present study showed that RET outcomes for traumatized immature non-vital teeth with A-PRF and PRF as scaffolds presented with similar clinical and radiographic findings at the end of 12 months of follow-up. A-PRF showed better root thickness than root length and PRF showed better root length than root thickness. Since gain in root thickness is crucial in immature teeth, A-PRF as a scaffold could be considered over PRF.

References

1. Trope M. Treatment of the immature tooth with a non-vital pulp and apical periodontitis. *Dent Clin North Am.* 2010;54:313-324.

2. Hargreaves KM, Diogenes A, Teixeira FB. Treatment Options: Biological Basis of Regenerative Endodontic Procedures. *J Endod.* 2013;3:S30-43.
3. Lin J, Zeng Q, Wei X, Zhao W, Cui M, Gu J, et al. Regenerative Endodontics Versus Apexification in Immature Permanent Teeth with Apical Periodontitis: A Prospective Randomized Controlled Study. *J Endod.* 2017;43:1821-7.
4. Diogenes A, Ruparel N. Regenerative Endodontic Procedures: Clinical Outcomes. *Dent Clin N Am.* 2017;61:111-125.
5. Chen M, Chen K, Chen C, Tayebaty F, Rosenberg P, Lin L. Responses of immature permanent teeth with infected necrotic pulp tissue and apical periodontitis/abscess to revascularization procedures. *Int Endod J.* 2012;45:294-305.
6. Murray PE, Garcia-Godoy F, Hargreaves KM. Regenerative endodontics: a review of current status and a call for action. *J Endod.* 2007;33:377-90.
7. Nosrat A, Kim J, Verma P, Chand P. Tissue engineering considerations in dental pulp regeneration. *Iran Endod J* 2014;9:30-9.
8. Bakhtiar H, Esmaili S, Fakhr Tabatabayi SF, Ellini MR, Nekoo-far MH, Dummer PM. Second-generation Platelet Concentrate (Platelet-rich Fibrin) as a Scaffold in Regenerative Endodontics: A Case Series. *J Endod.* 2017;43:401-8.
9. Narang I, Mittal N, Mishra N. A comparative evaluation of the blood clot, platelet-rich plasma, and platelet-rich fibrin in regeneration of necrotic immature permanent teeth: A clinical study. *Contemp Clin Dent.* 2015;6:63-8.
10. Choukroun J, Ghanaati S. Introducing the Low-Speed Centrifugation Concept. In: Miron RJ and Choukroun J (ed) *Platelet Rich Fibrin in Regenerative Dentistry: Biological Background and Clinical Indications.* Wiley Blackwell; 2017.p. 33-46.
11. Ghanaati S, Booms P, Orłowska A, Kubesch A, Lorenz J, Rutkowski J, et al. Advanced platelet-rich fibrin: a new concept for cell-based tissue engineering by means of inflammatory cells. *J Oral Implantol.* 2014;40:679-89.
12. Soto-Peñaloza D, Peñarrocha-Diago M, Cervera-Ballester J, Peñarrocha-Diago M, Tarazona-Alvarez B, Peñarrocha-Oltra D. Pain and quality of life after endodontic surgery with or without advanced platelet-rich fibrin membrane application: a randomized clinical trial. *Clin Oral Investig.* 2020;24:1727-38.
13. AAE Clinical Considerations for a Regenerative Procedure (2016). American Association of Endodontists. <https://www.aae.org/specialty/wp-content/uploads/sites/2/2017/06/currentregenerativeendodontic-considerations.pdf>. Accessed 2 Dec 2017
14. Saoud TM, Zaazou A, Nabil A, Moussa S, Lin LM, Gibbs JL. Clinical and radiographic outcomes of traumatized immature permanent necrotic teeth after revascularization/revitalization therapy. *J Endod.* 2014;40:1946-52.
15. Sabrah AH, Yassen GH, Liu WC, Goebel WS, Gregory RL, Platt JA. The effect of diluted triple and double antibiotic pastes on dental pulp stem cells and established *Enterococcus faecalis* biofilm. *Clin Oral Investig* 2015;19:2059-66.
16. Orstavik D, Kerekes K, Eriksen H. The periapical index: a scoring system for radiographic assessment of apical periodontitis. *Dent Traumatol.* 1986;2:20-34.
17. Friedman S, Mor C. The success of endodontic therapy--healing and functionality. *J Calif Dent Assoc.* 2004;32:493-503.
18. Bose R, Nummikoski P, Hargreaves K. A retrospective evaluation of radiographic outcomes in immature teeth with necrotic root canal systems treated with regenerative endodontic procedures. *J Endod.* 2009;35:1343-9.
19. Jeeruphan T, Jantarat J, Yanpiset K, Suwannapan L, Khewsawai P, Hargreaves KM. Mahidol study 1: comparison of radiographic and survival outcomes of immature teeth treated with either regenerative endodontic or apexification methods: a retrospective study. *J Endod.* 2012;38:1330-36.
20. Alobaid AS, Cortes LM, Lo J, Nguyen TT, Albert J, Abu-Melha AS, et al. Radiographic and clinical outcomes of the treatment of immature permanent teeth by revascularization or apexification: a pilot retrospective cohort study. *J Endod.* 2014;40:1063-70.
21. Sutam N, Jantarat J, Ongchavalit L, Sutimuntanakul S, Hargreaves K. A Comparison of 3 Quantitative Radiographic Measurement Methods for Root Development Measurement in Regenerative Endodontic Procedures. *J Endod.* 2018;44:1665-70.
22. Nazzal H, Kenny K, Altimimi A, Kang J, Duggal MS. A prospective clinical study of regenerative endodontic treatment of traumatised immature teeth with necrotic pulps using bi-antibiotic paste. *Int Endod J.* 2018;51:e204-15.
23. Zhou R, Wang Y, Chen Y, Chen S, Lyu H, Cai Z, Huang X. Radiographic, Histologic, and Biomechanical Evaluation of Combined Application of Platelet-rich Fibrin with Blood Clot in Regenerative Endodontics. *J Endod.* 2017;42:2033-39.
24. Jiang X, Liu H, Peng C. Clinical and radiographic assessment of the efficacy of a collagen membrane in regenerative endodontics: a randomized, controlled clinical trial. *J Endod.* 2017;43:1465-1471.
25. Ulusoy AT, Turedi I, Cimen M, Cehreli ZC. Evaluation of blood clot, platelet-rich plasma, platelet-rich fibrin, and platelet pellet as scaffolds in regenerative endodontic treatment: a prospective randomized trial. *J Endod.* 2019;45:560-6.
26. Kobayashi E, Flückiger L, Fujioka-Kobayashi M, Sawada K, Sculean A, Schaller B, et al. Comparative release of growth factors from PRP, PRF, and advanced-PRF. *Clin Oral Investig.* 2016;20:2353-60.
27. Masuki H, Okudera T, Watanebe T, Suzuki M, Nishiyama K, Okudera H, et al. Growth factor and pro-inflammatory cytokine contents in platelet-rich plasma (PRP), plasma rich in growth factors (PRGF), advanced platelet-rich fibrin (A-PRF), and concentrated growth factors (CGF). *Int J Implant Dent.* 2016;2:1-6.
28. Wang Y, Cox MK, Coricor G, MacDougall M, Serra R. Inactivation of Tgfb2 in Osterix-Cre expressing dental mesenchyme disrupts molar root formation. *Dev. Biol.* 2013;372:27-36.
29. Zhang W, Zhang X, Ling J, Wei X, Jian Y. Osteo-/odontogenic differentiation of BMP2 and VEGF gene-co-transfected human stem cells from apical papilla. *Mol Med Rep.* 2016;3:3747-54.
30. Nazzal H, Ainscough S, Kang J, Duggal MS. Revitalisation endodontic treatment of traumatised immature teeth: a prospective long-term clinical study. *Eur Arch Paediatr Dent.* 2019;6:1-10.
31. Kontakiotis E, Filippatos C, Tzanetakis G, Agrafioti A. Regenerative Endodontic Therapy: A Data Analysis of Clinical Protocols. *J Endod.* 2015;41:146-154.
32. Nagata JY, de Almeida Gomes BP, Lima TF, Murakami LS, de Faria DE, Campos GR, et al. Traumatized immature teeth treated with 2 protocols of pulp revascularization. *J Endod.* 2014;40:606-12.
33. Estefan BS, El Batouty KM, Nagy MM, Diogenes A. Influence of age and apical diameter on the success of endodontic regeneration procedures. *J Endod* 2016;11:1620-25
34. Yajamanya S, Chatterjee A, Babu C, Karunanithi D. Fibrin network pattern changes of platelet-rich fibrin in young versus old age group of individuals: A cell block cytology study. *J Indian Soc Periodontol* 2016;20:151-156.
35. Zeng Q, Nguyen S, Zhang H, Chebrolu HP, Alzebeid D, Badi MA, et al. Release of growth factors into root canal by irrigations in regenerative endodontics. *J Endod.* 2016;42:1760-66.
36. Huang X, Zhang J, Huang C, Wang Y, Pei D. Effect of intracanal dentine wettability on human dental pulp cell attachment. *J Endod* 2012;45:346-53.
37. Galler KM, D'Souza RN, Federlin M, Cavender AC, Hartgerink JD, Hecker S, et al. Dentin conditioning codetermines cell fate in regenerative endodontics. *J Endod.* 2011;37:1536-41.
38. Song M, Cao Y, Shin SJ, Shon WJ, Chugal N, Kim RH, et al. Revascularization-associated intracanal calcification: assessment of prevalence and contributing factors. *J Endod* 2017;43:2025-33.
39. Wattanapakkavong K, Srisuwan T. Release of transforming growth factor beta 1 from human tooth dentin after application of either Pro-Root MTA or Biodentine as a coronal barrier. *J Endod* 2019;45:701-5.

Acknowledgement

The authors wish to thank post-graduate students of Department of Conservative Dentistry and Endodontics and, Department of Paediatric Dentistry, JSS Academy of Higher Education and Research, for

providing cases for this study. We also thank Dr. Lancy D'Souza, Associate Professor, University of Mysore, for his statistical advice.

Ethics

All procedures performed in the present study were in accordance with the ethical standards of the institutional research committee (Protocol ID: JSS/DCH/IEC/MD-17/2017-2018) and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Source of funding

Funding: This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors

Authors' contributions

Veena Jayadevan, Paras Mull Gehlot and Subbarao V Madhunapantula contributed to the study conception and design. Material preparation and data collection were performed by Veena Jayadevan, Paras Mull Gehlot, Vinutha Manjunath and Jyothi S L. Data analysis were performed by Subbarao V Madhunapantula, Paras Mull Gehlot, Vinutha Manjunath and Jyothi S L. The original draft of the manuscript was written by Veena Jayadevan and Paras Mull Gehlot and, all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Conflict of Interest

Veena Jayadevan declares that she has no conflict of interest.

Paras Mull Gehlot declares that he has no conflict of interest.

Vinutha Manjunath declares that she has no conflict of interest.

Subbarao V Madhunapantula declares that he has no conflict of interest.

Jyothi Swandenahalli Lakshmikanth declares that she has no conflict of interest.